

Problem 280: Planetary Weight Converter

Difficulty: Easy

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Problem Background

Have you ever wondered how much you would weigh on another planet? Weight is not the same everywhere in the universe because it depends on gravity, which is the force that pulls objects toward the surface of a planet. Different planets have different amounts of gravity because of their size and mass.

Problem Description

You will write a program to calculate how much a person would weigh on each of the other planets in the solar system. You will be given several different weights of people on Earth, and must use the provided gravitational ratios to calculate their weight on the other planets. Display the weights on the other planets rounded to the nearest tenths place; be sure to check the information in the Reference Materials on how to correctly round numbers.

Planet	Gravity Ratio (Relative to Earth)
Mercury	0.377
Venus	0.905
Mars	0.379
Jupiter	2.528
Saturn	1.065
Uranus	0.886
Neptune	1.137

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include a positive integer representing the weight of the person on Earth (in Newtons, N).

```
2
72
100
```

Sample Output

For each test case, your program must print a line for each planet in the solar system in the order provided below. On each line, print the name of the planet, a colon and a space, then the weight of the person on that planet in N, rounded to the nearest tenth using the method provided in the Reference Materials. Include any trailing zeros when printing the weights.

- Mercury

- Venus
- Earth
- Mars
- Jupiter
- Saturn
- Uranus
- Neptune

Mercury: 27.1

Venus: 65.2

Earth: 72.0

Mars: 27.3

Jupiter: 182.0

Saturn: 76.7

Uranus: 63.8

Neptune: 81.9

Mercury: 37.7

Venus: 90.5

Earth: 100.0

Mars: 37.9

Jupiter: 252.8

Saturn: 106.5

Uranus: 88.6

Neptune: 113.7